1 Solution — GIAM 1.5 Problem 4

1.1 Problem

Trace through the Euclidean algorithm with inputs $a = 3731$ and $b = 2730$. Each time the assignment statement that calls the division algorithm is encountered, write out the expression $a = qb + r$ (with the actual values involved).

1.2 The Algorithm

The Section 1.5 pseudocode for *Euclidean*:

Label 1.
Let $(q, r) = \text{Division}(a, b)$.
If $r = 0$ **then Return** b.
Let $a = b$.
Let $b = r$.
Goto 1.

The assignment that calls the division algorithm is **Let** $(q, r) = \text{Division}(a, b)$. Each time it runs, Division returns the unique q, r with $a = qb + r$ and $0 \le r < b$. We write out that equation, then — if $r \neq 0$ — shift $(a, b) \leftarrow (b, r)$ and loop.

1.3 The Trace ($a = 3731$, $b = 2730$)

Each line below is one execution of the Division call:

$$
\begin{aligned}
3731 &= 1 \cdot 2730 + 1001, \\
2730 &= 2 \cdot 1001 + 728, \\
1001 &= 1 \cdot 728 + 273, \\
728 &= 2 \cdot 273 + 182, \\
273 &= 1 \cdot 182 + 91, \\
182 &= 2 \cdot 91 + 0. \qquad \Leftarrow\ r = 0,\ \textbf{Return}\ b = 91.
\end{aligned}
$$

The remainder hits 0 on the sixth call, so the algorithm returns the last divisor:

$$\boxed{\gcd(3731, 2730) = 91, \quad \text{found in 6 calls to Division.}}$$

1.4 The Remainder Cascade

Notice how each line's divisor and remainder become the next line's dividend and divisor — the pair (a, b) slides down to (b, r) every step:

Each call Division(a, b) returns (q, r) with a = qb + r. The remainder r becomes the next divisor: (a, b) → (b, r). Stop when r = 0; the last divisor is the gcd.

call #	a = q · b + r	next (a, b)
1	3731 = 1 · 2730 + 1001	(2730, 1001)
2	2730 = 2 · 1001 + 728	(1001, 728)
3	1001 = 1 · 728 + 273	(728, 273)
4	728 = 2 · 273 + 182	(273, 182)
5	273 = 1 · 182 + 91	(182, 91)
6	182 = 2 · 91 + 0	r = 0 → stop

The division algorithm is called **6 times**. Last nonzero divisor → **gcd(3731, 2730) = 91**.

Check: 91 = 7·13, 3731 = 7·13·41, 2730 = 2·3·5·7·13 — the shared part is 7·13 = 91.

Figure 1.1: Six calls to Division. Row 1: 3731 = 1·2730 + 1001; the divisor 2730 and remainder 1001 become the next (a,b). Each subsequent row shifts (a,b) → (b,r): 2730=2·1001+728, 1001=1·728+273, 728=2·273+182, 273=1·182+91, 182=2·91+0. The remainder reaching 0 on call 6 stops the loop; the last divisor 91 is the gcd. Check: 91 = 7·13, dividing both 3731 = 7·13·41 and 2730 = 2·3·5·7·13.

1.5 Remark — Why the Last Nonzero Remainder Is the gcd

The algorithm works because of one key fact: $\gcd(a,b) = \gcd(b,r)$ whenever $a = qb + r$. Any number dividing both a and b also divides $r = a - qb$; conversely any number dividing b and r divides $a = qb + r$. So the *set* of common divisors — and hence the greatest one — is unchanged at every step. Tracking the $\gcd$ down the cascade:

$$\gcd(3731, 2730) = \gcd(2730, 1001) = \gcd(1001, 728) = \cdots = \gcd(182, 91) = \gcd(91, 0) = 91.$$

When the remainder finally reaches 0, the pair is $(91, 0)$, and $\gcd(91, 0) = 91$ because everything divides 0 — so 91 is its own gcd with 0. That is why the *last nonzero remainder* (equivalently, the last divisor) is the answer.

1.6 Remark — Counting the Division Calls

The question highlights the $\mathsf{Division}$ call specifically because that is the algorithm's real unit of work — and here it runs exactly 6 times. Note this is *one* assignment per loop pass, in contrast to the repeated-subtraction division of Problem 1 which did many cheap assignments. Each Euclidean step does a single (expensive) division but slashes the numbers dramatically: $3731 \to 2730 \to 1001 \to 728 \to 273 \to 182 \to 91$. The number of steps grows only like the number of *digits* of the inputs, which is why Euclid is so fast — six divisions here versus the hundreds of subtractions a naïve approach would need.

1.7 Remark — The Quotients and a Glimpse of Continued Fractions

The quotients produced, in order, were

$$q = 1,\ 2,\ 1,\ 2,\ 1,\ 2.$$

These are not throwaway by-products: read top to bottom they are exactly the **continued-fraction expansion** of a/b,

$$\frac{3731}{2730} = 1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{2}}}}}.$$

So a single run of the Euclidean algorithm simultaneously delivers the $\gcd$ *and* the continued fraction of the ratio — a small preview of how much structure is packed into this short loop.

1.8 Remark — Contrast With the Factoring Method

We could instead have factored: $3731 = 7 \cdot 13 \cdot 41$ and $2730 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 13$, whose shared part is $7 \cdot 13 = 91$ — matching the boxed answer, and illustrating the min-exponent rule of Problem 3. But finding those factorizations is *harder* than running Euclid: spotting that $3731 = 7 \cdot 13 \cdot 41$ takes real work, whereas the six divisions above are mechanical. This is the recurring lesson of the section — Euclid computes the same $\gcd$ the factorization *describes*, but does so without the (expensive) detour through factoring.